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RESEARCH

MHC-Segnet: Mamba–Hadamard collaboration segmentation network for multimodal MRI brain tumor

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Abstract

Brain tumor segmentation from multimodal magnetic resonance imaging is essential for computer-aided diagnosis and therapy formulation. There are still a number of obstacles in the accurate segmentation for the computer-assisted application. CNN-based encoder faces challenges in capturing long-range dependencies and lacks robustness when using skip connections, which leads to loss of details and affects segmentation accuracy. Transformer-based encoder can obtain global context, but it is limited by computational constraints when processing high-resolution medical images. To address these issues, we propose MHC-SegNet, a novel Mamba–Hadamard collaboration segmentation network that aims to more efficiently capture global patterns across modalities. We design a collaboration encoder–decoder with scale-aware Mamba encoder and triple-axis group Hadamard product to capture long-range dependencies and extract multi-perspective 3D spatial features with computationally efficient. In addition, scale-aware feature refinement layers are also proposed to represent and aggregate multi-scale information in the collaboration encoder, and a hierarchical fusion skip module is designed to address the loss of detail caused by vanilla skip connections. The proposed modules enable the network to achieve a comprehensive understanding of the input data, and accurately identify tumor features. Experimental results demonstrate that our model achieves the best performance compared to recent state-of-the-art models in segmentation accuracy.

Keywords Brain tumor segmentation · Multimodal MRI · Mamba · Hadamard product

1 Introduction

Brain tumor segmentation is vital for automated biomedical analysis [1, 2] as it provides critical insights for accurate diagnosis and preparation of therapy. In recent years, segmentation networks-based convolutional neural network (CNN) [3] or Transformer are widely used and have demonstrated outstanding performance in semantic segmentation. Recent studies [4–6] in computer-aided brain tumor segmentation have made notable progress.

The CNNs are highly regarded in segmentation for their strong inductive bias and effective feature extraction capabilities, excelling at capturing local features. Transformer networks [7] leverage global attention mechanisms to indicate strong capabilities in modeling global features and handling complex medical images, helping models understand global information and complex relationships. However, CNN retrieves feature using local receptive fields, neglecting global context and long-range relationships. This inhibits their ability to represent features in complicated, high-dimensional medical images. Transformer network can simulate global self-attention traits [8, 9]; however, they are limited in processing MRI tumor imaging due to computational resource and memory limitations.

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Existing encoders [10–12] face challenges in the modeling limitations of remote information caused by local convolutional operations and the high computational overhead of transformers with insufficient spatial perception ability. In these circumstances, enhancing the computational efficiency and spatial awareness in long-range dependency modeling is particularly imperative. Mamba [13] has demonstrated powerful capabilities in handling long-range dependency relationships and improving computational efficiency in recent research, making it an effective solution for alleviating the computational burden of 3D medical image segmentation. Therefore, it is worthy to explore methods that can effectively capture multi-view data [14] and enhance multimodal feature representations.

In our work, we design shallow SME blocks and deep TGHE blocks in the encoder to effectively handle high-dimensional data. For the SME block, Mamba can efficiently model long-range dependencies to capture global information. Moreover, the SFR layers are introduced to fuse multi-scale information [15]. For the TGHE block, we design TGHP layers to achieve 3D features' extraction and fusion, capturing multi-perspective spatial information. In addition, we conduct extensive experiments on the BraTS2019 dataset, which has become a reliable choice for current brain tumor segmentation research by standardizing segmentation standards and ensuring quality and comparability of research results. The results show that our method achieves excellent performance and higher segmentation accuracy compared to recent SOTA methods. The main contributions of the work can be summarized as follows:

- We propose a novel MHC-SegNet for the multimodal brain tumor segmentation, which is a robust encoder–decoder network to achieve efficient and high 3D segmentation performance, surpassing SOTA methods with the 0.51% improvement.
- We propose a collaboration encoder consisting of SME blocks and TGHE blocks, which utilizes SFR and Mamba to improve the computational efficiency and spatial awareness in long-range dependency modeling, and the introduced TGHP is used to extract multi-perspective drawing spatial information with the low computing cost. The experimental results demonstrate that SME and THGE blocks improve the performance improvements of 1.36% and 1.30%, respectively.
- We propose the HFSkip module, which captures high-dimensional and low-dimensional features by fusing sub-coding blocks at each level. This design reduces detail loss in skip connections and preserves richer global information, achieving a 1.29% performance improvement compared to the baseline.

2 Related work

2.1 BraTS and CNN-based segmentation network

The brain tumor segmentation (BraTS) [16–18] is a comprehensive collection of multimodal magnetic resonance imaging scanning dataset designed for the evaluation of brain tumor segmentation algorithms. This dataset brings together research teams from around the world to utilize various advanced algorithms and deep learning techniques segmenting brain tumors such as gliomas. These efforts are fundamentally important for improving the diagnosis, treatment planning, and prognosis assessment of brain tumors.

U-Net [19, 20] has significantly advanced the CNN-based networks [21–23] in 2D and 3D medical image segmentation [24, 25]. To achieve volumetric segmentation of brain tumors, researchers employ multiple strategies, including using multi-planar architectures to combine slices from different perspectives. These approaches [26–28], referred to as the 2.5D method, have been extensively explored. In addition, Zhang et al. developed 3D method that directly utilizes complete volumetric images or stack 2D images to form a 3D representation, which enables better capture of the three-dimensional structural information of tumors [29].

Relative to prevailing methodologies [30, 31], our framework amalgamates their distinct advantages and introduces a triaxial Hadamard p-scanning and a novel integration of skip connections, further enhancing the accuracy and reliability of brain tumor segmentation to achieve high-quality fusion.

2.2 Transformers and state space models

Transformers exhibit remarkable performance in image processing [32–34], capturing long-range information and modeling global contextual information through their self-attention mechanism. Vision transformer (ViT) [35] treats splitted image as sequential inputs, achieving success in large-scale training and sparking extensive research on Transformer [7]. Models that combine CNN and transformers, such as TransBTS [36] and 3DTransUnet [37], leverage the local feature extraction capabilities of convolutions and the global feature modeling advantages of Transformer. 3DUX-Net [38] incorporates large kernel convolution blocks to model global features using the large kernel convolution method. ConvNeXt [39] presents a pure convolutional model simulating Transformer structures within CNNs. SwinUNETR [40] integrates SwinTransformer [41] into U-Structure and enhances decoder efficacy via multi-resolution strategies.

Transformer architecture is known for its effectiveness, it has the $O(n^2)$ complexity that incurs significant computational overhead in medical segmentation. Recently, selective state spaces from state space models (SSMs) are used across

image, audio, and text processing domains. Mamba [13] integrates these selective SSMs into a simplified end-to-end neural network architecture, relieves convolutional neural networks' modeling constraints, and offers advanced modeling capabilities akin to Transformer through global receptive fields and dynamic weighting. VMamba [42] expands image blocks in the horizontal and vertical dimensions of images and bidirectionally scans in the two directions. Vision Mamba [43] enhances the performance of visual Mamba by integrating spatial information with position embeddings. MambaND [44] extends the Mamba to be able to process multidimensional data including images and videos. Segmamba [45] captures multi-scale features, and uses Mamba to model the global information of high-dimensional features. In contrast to these approaches, we propose SME block that integrates SFR prefix layer and Mamba layer, adopting sweep scan to facilitate seamless feature aggregation at different scales.

3 Mamba–Hadamard collaboration segmentation network

3.1 Architecture

The overall framework of our approach, depicted in Fig. 1, follows the encoder–decoder [33] design commonly used in many brain tumor segmentation approaches [46–48]. Our framework comprises the collaboration encoder that combines two shallow sub-encoders consisting of SME blocks and two deep sub-encoders consisting of TGHE blocks, a lightweight decoder, and the HFSkip module.

Given an input volume with a shape of $M \times (L)^3$, where M represents the number of modalities, we initially transform it into a 3D medical image with a shape of $M \times W \times H \times D$ through Stem module. Then, we incorporate two shallow sub-encoders forming a collaboration encoder [49, 50], each comprising N_0 and N_1 SME blocks to facilitate feature fusion across multiple views through the fusion of different scales. The features are then fused using two sub-encoders made up of N_2 and N_3 TGHE blocks to perform triple-axis spatial feature fusion. Here, $N_x \in \{N_0, N_1, N_2, N_3\}$ denotes the number of blocks in each sub-encoder. The HFSkip module, a crucial component of our network, is positioned between the encoder and decoder. It plays a vital role in fusing multidimensional features and mitigating the loss of feature information. By establishing skip connections between corresponding encoding and decoding layers, the HFSkip module facilitates the seamless flow of information from the encoder to the decoder. This integration of multidimensional features enhances the overall feature representation and minimizes the reduction of crucial information during the encoding-decoding process. In the end, the decoder is made up of four decoding layers, each of which has a TransConv3D layer and a ResConv3D layer. To get the final segmentation output, the features are decoded and then sent through a TransConv3D Head.

3.2 Scale-aware Mamba encoder

In the SME, the input feature is initially processed by the SFR module, which performs feature extraction and integration through multi-scale Conv3D, enhancing spatial details and contextual information. These features are then passed to the Mamba layer, which captures global dependencies and temporal dynamics by flattening, layer normalization, and state space models, enriching the contextual representation of the features. Lastly, the expansion–compression mechanism, denoted by EC layer in Fig. 1, performs nonlinear transformation and channel dimension expansion of the features, providing richer feature representations for subsequent encoding stages.

3.2.1 Scale-aware feature refinement

The 3D features are flattened into the 1D sequence by the Mamba layer, which then models the feature relationships. Therefore, prior to the Mamba layer, in order to capture the spatial feature, SFR layer is designed to integrate features from multiple scales.

To begin with, the SFR applies Conv3D layers with different kernel sizes, 1, 3, and 5, denoted as $C^{k\times k\times k}$, to process the input feature maps F. This enables the extraction of features at different spatial resolutions, capturing both local details and contextual information. The resulting feature maps are then activated using the Gaussian error linear unit (GELU) activation function, which introduces nonlinearity and enhances the expressiveness of the network. In particular, another Conv3D layer with a kernel size of 1 is utilized to further refine the high-level features extracted by the Conv3D layer with a kernel size of 5. The initial process is defined as follows:

$$\phi_k(F) = \mathrm{GELU}(GN(C^{k\times k\times k}(F))), \tag{1}$$

where $\phi_k(F)$ represents the feature map obtained by convolving F with the kernel size of k, followed by group normalization, denoted as $GN(\cdot)$ and activation using the GELU, denoted as $\mathrm{GELU}(\cdot)$. The final output feature map $\Phi(F)$ is obtained as:

$$\Phi(F) = C^{1\times 1\times 1}(\phi_1(F) \oplus \phi_3(F) \oplus \phi_1(\phi_5(F))), \tag{2}$$

where $\phi_1(F)$, $\phi_3(F)$, and $\phi_1(\phi_5(F))$ denote the element-wise summation of the feature maps. $C^{1\times 1\times 1}$ denotes a

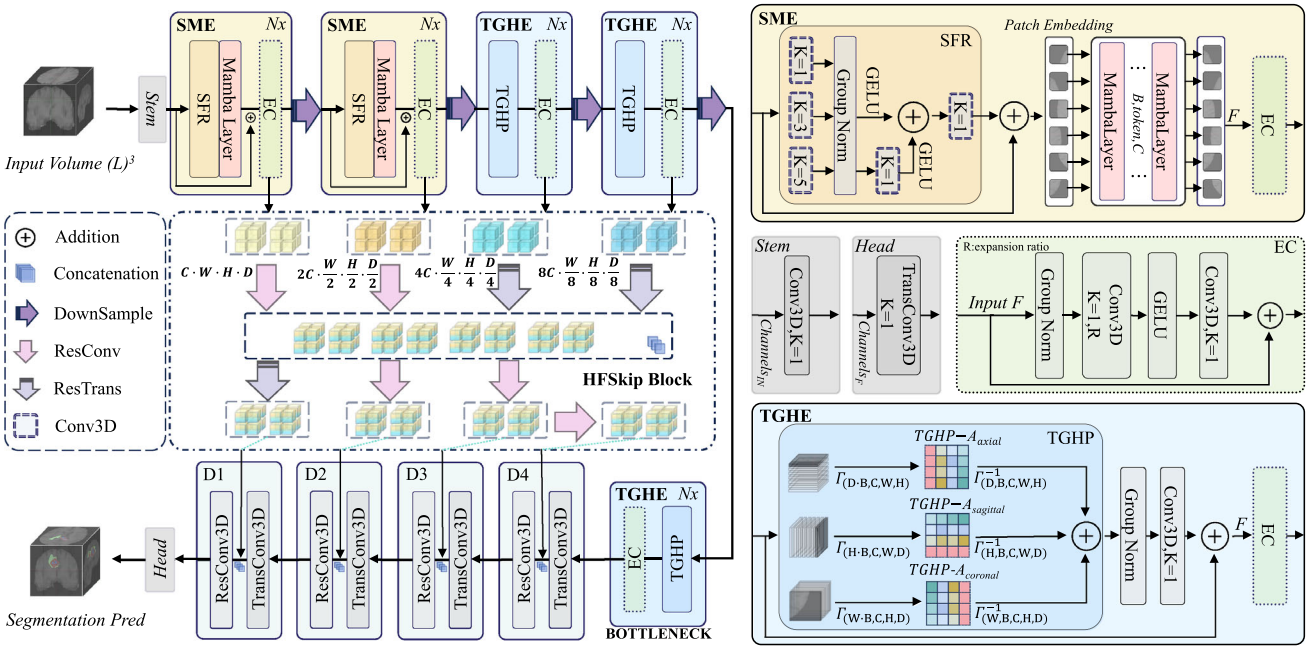


Fig. 1 The overall framework of MHC-SegNet. The model incorporates SME and TGHE modules for hierarchical encoding of multimodal 3D inputs. The SME leverages the SFR head to capture multi-scale features while employing Mamba sequential modeling to extract contextual information. The TGHE utilizes tri-view Hadamard product to enhance spatial alignment, and HFSkip blocks ensure effective fusion of multilevel encoder features within the decoder

Conv3D layer with a kernel size of 1. $\oplus$ denotes the element-wise summation operation.

3.2.2 Mamba layer

The Mamba layer is a pivotal component in the SME, designed to model global dependencies within feature maps. As shown in Fig. 2, it initiates the process by flattening the feature voxel into a 1D sequence ψ'_f, which is then normalized using layer normalization to stabilize and accelerate the learning process. We perform sweep-scan planning of the feature voxel from the axial direction to scan the ψ_f and its reversed sequence ψ'_f using the idea of bidirectional mamba, which allows the Mamba layer to capture contextual information in both forward and backward directions, resulting in the improved understanding of global dependencies and the enhanced representation of long-range dependencies. The process of the Mamba layer can be expressed as follows:

$$\Psi_m = \text{Mamba}(LN(\psi_f), LN(\psi'_f)), \tag{3}$$

where ψ_f represents the flattened feature sequence, ψ'_f denotes reversed ψ_f, $LN(\cdot)$ denotes layer normalization, and Ψ_m is the output sequence from the Mamba layer that encapsulates enriched contextual information. Then, the output sequence from the Mamba layer is reshaped back to the original 3D volume to ensure compatibility with subsequent encoder.

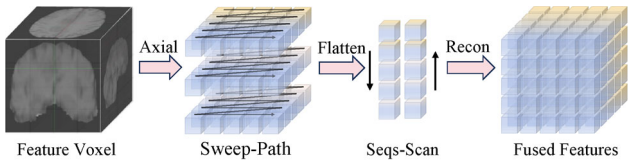


Fig. 2 The workflow of Mamba layer. As the core component of SME, the Mamba layer obtains global dependencies by flattening feature voxels into a sequence ψ'_f and normalizing it with layer normalization to stabilize learning

3.3 Triple-axis group Hadamard encoder

As a solution to improve computing efficiency, we provide the TGHP layer, a near-linear complexity strategy that attempts to extract information from many viewpoints [51], as a significant component of TGHE. For the Hadamard product (HP) process, given an input tensor x and a randomly initialized learnable tensor p, we first resize p to match the size of x using bilinear interpolation. Then, we apply depth-wise separable convolution (DWConv) on p, followed by an HP between x and p to obtain the output. This process allows the integration of information from different views.

Nevertheless, simple HP is insufficient to extract information from multiple perspectives, resulting in the loss of

spatial information and unable to undertake 3D volume tasks, leading to unsatisfactory results. Inspired by the multi-head mode in multi-head self-attention mechanism, we introduce TGHP-A in our proposed method. TGHP-A simplifies the information extraction process by dividing the input into four groups along the channel dimension. It performs HP on three groups including the height–width, channel–height, and channel–width axes, and utilizes DWConv on the fourth group. The groups are then concatenated, and another DWConv is applied for integration. The DWConv kernel sizes are consistently set to 3. In total, the TGHP-A processing can be expressed as follows:

$$\gamma(x) = \sum_{i,j}^{3,J} x_i \odot DW(BI(P_j)) + DW(x_4),\ j \in (xy, zx, zy), \tag{4}$$

where x_i represents the i-th group of the input tensor x, P_j is the j-th learnable tensor, $\odot$ denotes the HP, DW represents depth-wise separable convolution, and BI denotes bilinear interpolation.

Significantly, we split the 3D input X of shape $C \times W \times H \times D$ along three axes, resulting in D axial slices of shape $C \times W \times H$ denoted as X_a^{-1}, H sagittal slices of shape $C \times W \times D$ denoted as X_s^{-1}, and W coronal slices of shape $C \times H \times D$ denoted as X_c^{-1}. The aggregation of these slices is described as follows:

$$\Gamma(X) = C^{1\times1\times1}\left(GN\left(\sum_{i}^{l}\gamma\left(X_i^{-1}\right)\right)\right) + X',\ l \in [a, s, c], \tag{5}$$

where $\Gamma(X)$ represents the output of the TGHP module, $C^{1\times1\times1}$ denotes a Conv3D with kernel size of 1, $GN(\cdot)$ stands for group normalization, and X' is the residual connection from the input tensor X. The summation $\sum_i^l \gamma(X_i^{-1})$ integrates the processed slices from the three axes. Similar to the SME, the expansion–compression mechanism is utilized as the final operation to enrich the feature representations of the sub-encoder. By splitting the input features along axes and aggregating processed slices, the TGHE enriches feature representations, resulting in improved performance for 3D volume tasks.

3.4 Hierarchical fusion skip

In order to address the limitation of vanilla skip connections in capturing the semantic information of each layer during the downsampling and transferring features effectively in the upsampling, we propose a novel HFSkip module, which can effectively integrate the feature information of different levels through multilevel feature fusion and skip connection, so as to improve the robustness and detail retention ability of the model. The compositions of the HFSkip module are shown in Fig. 3. Initially, we extract feature maps containing different resolutions from four different layers. The shapes of the feature maps S_i can be expressed as follows:

$$S_i = \left[2^i \times C, W/2^i, H/2^i, D/2^i\right], \tag{6}$$

where i represents the index of the feature map, C represents the initial number of channels, and W, H, D represents the width, height, and depth of the input feature. Each feature map is processed through scale normalization (SN) layer by utilizing specific convolutional blocks to facilitate hierarchical fusion. These convolutional blocks include downsampling, upsampling, and residual, so as to adapt to different layers of feature embedding. The processed feature maps are concatenated in the channel dimension to form a hierarchical fusion feature map, which is further fused through a series of convolutional blocks to generate a richer feature description and achieve higher accuracy and better detail retention ability. After that, the HFSkip module outputs four feature maps, the shapes of which can be expressed as:

$$S_i' = \begin{cases} SN'\left(FU\left(SN\left(S_i\right)\right)\right), & i \in \{1, 2, 3\} \\ SN'\left(S_3'\right), & i = 4 \end{cases}, \tag{7}$$

where S_i represents the input feature map at the i-th level of the HFSkip module, $SN(\cdot)$ denotes the scale normalization operation, $FU(\cdot)$ refers to the fusion operation, $SN'(\cdot)$ represents the inverse scale normalization operation, which restores the fused feature map to its original scale, and S_i' represents the output feature map at the i-th level of the HFSkip module after the fusion process. Lastly, these feature maps are, respectively, sent to the corresponding decoders for reconstructing the desired output.

3.5 Loss function

We employ a hybrid loss function [52], which combines binary cross-entropy (BCE) loss and Dice loss, to optimize model training for multimodal brain tumor segmentation tasks. Dice loss, L_{dice}, measures the overlap between the predicted segmentation and the ground truth segmentation to assess the quality of the segmentation [53]. BCE loss, L_{bce}, quantifies the difference between the predicted probability distribution and the true label distribution. In specific, the L_{dice} can be expressed as:

$$L_{dice} = 1 - \frac{2}{M}\sum_{m=1}^{M}\frac{\sum_{n=1}^{N} X_{n,m}Y_{n,m}}{\sum_{n=1}^{N} X_{n,m} + \sum_{n=1}^{N} Y_{n,m}}, \tag{8}$$

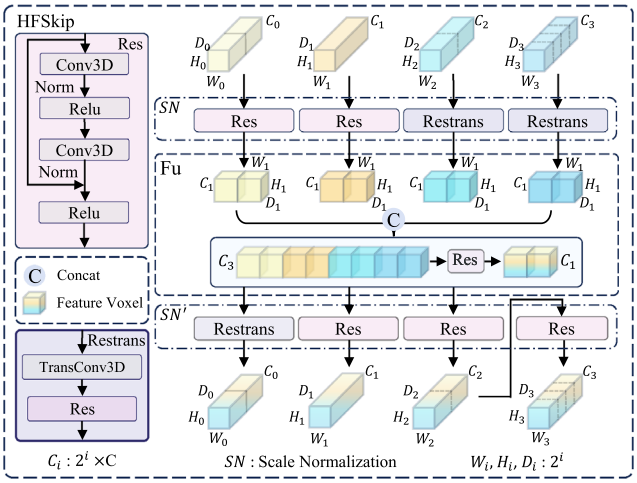


Fig. 3 The hierarchical fusion skip module. The layer integrates features from each encoder stages with decoder, employing convolutional scaling and skip connections to achieve multilevel information and enhance model robustness

where $Y_{n,m}$ and $X_{n,m}$ stand for the probability output and one-hot encoded ground truth for class m at voxel n, respectively; N represents the number of voxels and M the number of classes. And L_{bce} can be represented as:

$$L_{bce} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log \hat{y}_i + (1-y_i)\log\left(1-\hat{y}_i\right)\right], \quad (9)$$

where N is the total number of samples, and y_i and $\hat{y}_i$ represent the true probability and model output for sample i. Furthermore, to enhance pixel-level classification performance, we combine binary BCE loss and Dice loss into a comprehensive L_{bce_dice}, which can be expressed as:

$$L_{bce_dice} = L_{dice} + L_{bce}, \quad (10)$$

To further ensure the robustness of the model in different tumor regions, we separately calculate the losses L_{NCR}, L_{ED}, and L_{ET} for necrotic tumor core, peritumoral edema, and enhancing tumor using the L_{bce_dice}. The total loss function, L_{bwd}, can be represented as follows:

$$L_{bwd} = \lambda_1 * L_{NCR} + \lambda_2 * L_{ED} + \lambda_3 * L_{ET}, \quad (11)$$

where the weights of the loss function have been optimized, and their default values are $\lambda_1 = \lambda_2 = \lambda_3 = 1$.

Table 1 Quantitative evaluations on BraTS2019 validation set

Model	Avg.		subset1		subset2		subset3		subset4		subset5		Params(M)
	DSC	SDC	DSC	SDC	DSC	SDC	DSC	SDC	DSC	SDC	DSC	SDC	
3DUnet [58] (MICCAI,2016)	72.05	60.78	68.31	57.62	70.14	59.62	77.28	65.06	76.44	63.53	68.07	58.09	**16.32**
TransBTS [36] (MICCAI,2021)	80.94	71.35	79.35	68.90	76.15	63.50	83.22	**75.26**	85.87	77.45	80.09	71.64	32.99
nnUnet [59] (Nature Method,2021)	81.61	72.31	79.75	69.52	79.82	69.51	82.67	74.21	85.68	76.77	80.15	71.52	31.19
SwinUNETR [40] (MICCAI,2021)	81.73	72.82	79.47	69.91	**80.76**	**71.00**	82.66	74.20	85.51	77.09	80.27	71.91	62.19
TransUnet [37] (MIA,2024)	79.64	70.02	78.46	68.00	77.80	67.62	80.35	71.76	84.88	75.82	76.69	66.92	31.20
Segmamba [45] (MICCAI,2024)	82.26	73.10	80.28	70.19	80.56	69.99	83.21	74.58	86.19	**78.07**	81.05	72.69	67.42
Ours	**82.77**	**73.47**	**80.64**	**70.68**	80.57	69.90	**83.94**	74.72	**86.73**	77.97	**81.98**	**74.04**	21.63

The first two results are marked with bold and underlined

4 Experimental results and analyses

4.1 Datasets

We evaluate the proposed MHC-SegNet on the BraTS2019 dataset, which includes 335 training samples. All patient data is preprocessed to a common spatial resolution and undergone skull stripping. The four original modal volume sizes of each data are unified as $240 \times 240 \times 155$. In each volume, the brain tumor region is segmented into three main sub-regions: the enhancing tumor (ET) region, the tumor core (TC) region (including the enhancing tumor and necrotic regions), and the whole tumor (WT) region (including the tumor core and edema regions).

The experimental results on the BraTS2019 dataset comprehensively demonstrate the effectiveness and robustness of the proposed method in addressing complex brain tumor segmentation tasks [54, 55]. A fivefold cross-validation strategy is employed to ensure the stability and reliability of the results, with 80% of the data allocated for training and the remaining 20% used for validation in each fold.

4.2 Evaluation metrics

We utilize the Dice similarity coefficient (DSC) and surface Dice coefficient (SDC) as evaluation metrics, which are commonly used to assess the accuracy of segmentation. The DSC measures the overall segmentation quality, while the SDC evaluates the accuracy of the segmented surface. The DSC can be expressed as follows:

$$DSC(G, P) = \frac{2\sum_{i=1}^{I} G_i P_i}{\sum_{i=1}^{I} G_i + \sum_{i=1}^{I} P_i}, \tag{12}$$

where G_i and P_i, respectively, denote the true and predicted values for i-th voxel of the volume. The SDC, which represents the predicted and actual accuracy on the boundary, can be calculated as:

$$SDC(S_{b,c}, S'_{b,c}) = \frac{\left|S_{b,c} \cap S_{b,c}^{'(\tau)}\right| + \left|S_{b,c} \cap S_{b,c}^{'(\tau)}\right|}{\left|S_{b,c}\right| + \left|S'_{b,c}\right|}, \tag{13}$$

where $S_{b,c}$ and $S'_{b,c}$, respectively, represent the ground truth surface and predicted surface of class c under batch index b. $S_{b,c}^{'(\tau)}$ represents the predicted segmentation values that are less than or equal to the threshold [56] τ compared to the surface $S_{b,c}$. Similarly, $S_{b,c}^{(\tau)}$ represents the well-defined region computed with respect to the surface $S_{b,c}^{'(\tau)}$. The tolerance τ is set to a default value of 1 mm.

4.3 Implementation details

All models are developed utilizing Python 3.8 and PyTorch 1.2 frameworks in conjunction with MONAI 1.3 and Mamba-SSM 1.2 libraries. During the experiments, we train the models for 1000 epochs on four A100 GPUs, with a batch size of 2 per GPU. We use a cosine annealing learning rate schedule [57] to dynamically tune the learning rate of the optimizer, with the initial learning rate set to 1e−3 and weight decay to 3e−5, and use the SGD optimizer. The input 3D data with four modals is cropped to a uniform size of $128 \times 128 \times 128$. Random flip is applied to increase data diversity, and random intensity shift is used for data augmentation. To ensure the authenticity of the results, all experiments are conducted using the same initialization parameters and settings.

4.4 Quantitative evaluations

We evaluate our proposed method against multiple models for brain tumor segmentation, including 3DUnet [58], nnUnet [59], TransBTS [36], SwinUNETR [40], 3DTransUnet [37], and Segmamba [45], which are based on CNN and Transformer [7] architectures.

Table 1 presents the experimental results of all the models. Our proposed MHC-SegNet demonstrates superior performance compared to all other models, achieving the highest scores in both DSC and SDC metrics at 82.77% and 73.47%, respectively. These results underscore the notable advantages of our network in terms of segmentation accuracy and consistency.

While the 3D U-Net architecture has been considered revolutionary, its segmentation performance in the current medical image analysis field has shown some limitations. It is noteworthy that our proposed MHC-SegNet achieves best performance with the parameter count closest to that of the 3D U-Net, highlighting its comparative advancements.

4.5 Qualitative results

The qualitative results are shown in Fig. 4, where we present the visual plane images of different modalities for the more intuitive comparison. We specifically select the challenging axial plane clip, where each row represents the segmentation results of the network on different modalities. Most networks perform well in segmenting simple modalities, but struggle to accurately differentiate the necrotic regions and enhanced tumor areas in complex modalities, resulting in suboptimal segmentation outcomes. Our proposed method, MHC-SegNet, not only achieves precise segmentation of boundaries and surrounding tissues in simple modalities but also accurately localizes the tumor core in complex modalities. This leads to higher segmentation accuracy. Overall, our

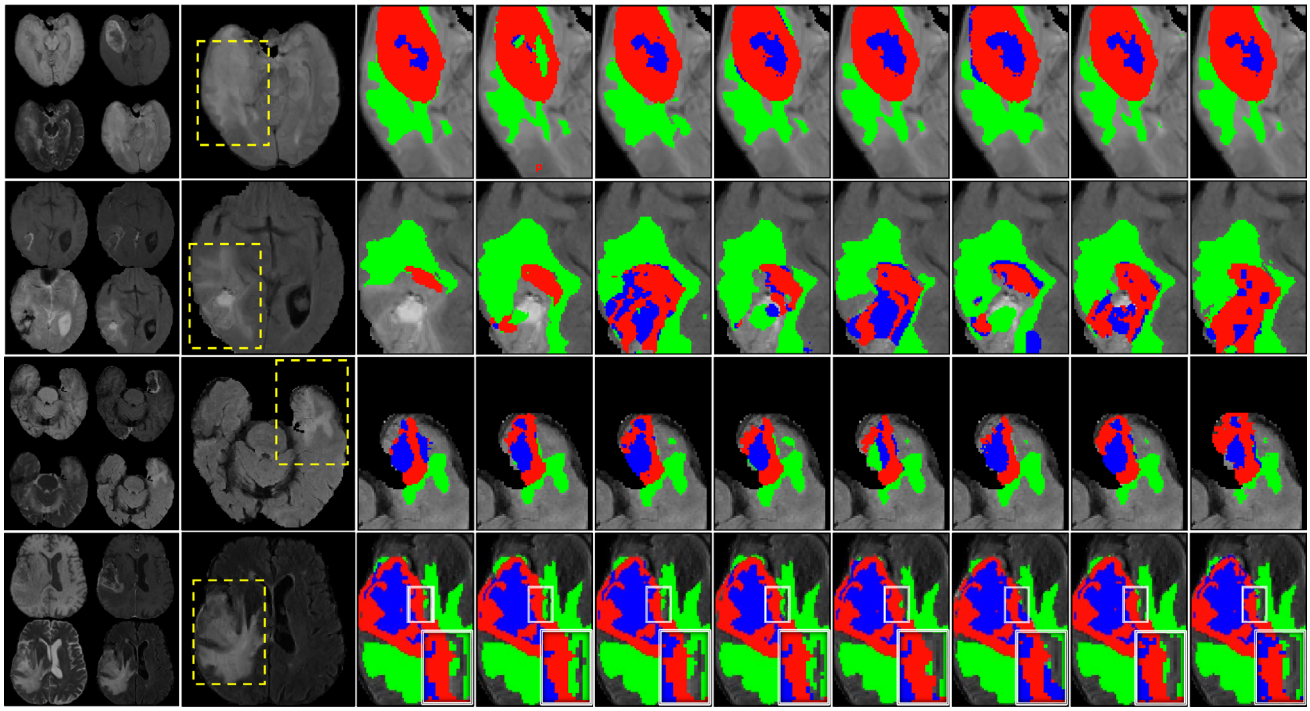


Fig. 4 Comparison of different methods with four difficulty levels. Each row represents a distinct challenge: average difficulty (non-connectivity in yellow regions and edge precision in blue areas), highest difficulty (general similarity), small-object segmentation, and detailed edge accuracy in simpler tasks. The blue and green regions represent the necrotic tumor core and peritumoral edema. The enhancing tumor is represented by the red region. The tumor core includes the combination of the red and blue regions. The enhancing whole tumor comprises the combination of the red, blue, and green regions

Table 2 Ablation studies of MHC-SegNet on BraTS2019

Components					Performance				
B	Mamba	SME	TGHP	HFSkip	ET	TC	WT	DSC	SDC
✓					70.40	82.97	88.43	80.59	70.82
✓	✓				71.95	84.29	89.22	81.81	70.98
✓		✓			71.99	<u>84.52</u>	89.35	<u>81.95</u>	71.10
✓			✓		71.94	84.12	89.62	81.89	71.36
✓				✓	<u>72.27</u>	83.64	<u>89.73</u>	81.88	<u>72.39</u>
✓	✓	✓	✓	✓	**73.79**	**84.63**	**89.91**	**82.77**	**73.47**

The first two results are marked with bold and underlined. SME denotes the combination of SFR and Mamba

method demonstrates superior performance in effectively utilizing modality features and capturing detailed information.

4.6 Ablation studies

We conduct a series of ablation experiments to validate the effectiveness of each module in the model and illustrate the fundamental design principles based on experiments using the BraTS2019 datasets. In this experiment, we design a baseline with an encoder based on the UNETR structure, utilizing MONAI's [60] unetrBasicBlock and supporting 3D volume input, and systematically integrate various modules to assess their incremental impact on performance. All experiments follow the implementation details mentioned above, with consistent preprocessing and post-processing.

We validate the performance by sequentially comparing the six models listed in Table 2, and present the quantitative results. The DSC and SDC for the ET, TC, and WT consistently exceed those of the baseline.

Specifically, comparing the baseline with the baseline model incorporating Mamba underscores the benefits of employing Mamba's bidirectional sweep scan for global sequence feature extraction. Adding SFR to the baseline model with SME further enhances relevant metrics, with

TC and DSC achieving a suboptimal value of 84.52% and 81.95%, respectively. The comparison between the baseline with and without TGHP reveals the efficacy of the multiaxial perspective in achieving DSC levels comparable to SME while further enhancing SDC. Incorporating HFSkip into the baseline demonstrates its effectiveness in refining edge features, leading to an advanced SDC of 72.39% within the module. The model combining SME, TGHP, and HFSkip with the baseline demonstrates its best performance, achieving global optimal levels for ET, TC, and DSC at 73.79%, 84.63%, and 82.77%, respectively.

The ablation studies demonstrate that our MHC-SegNet, integrating all technological innovations, achieves best performance comprehensively in metrics of DSC and SDC, exhibiting remarkable efficacy in both quantitative and qualitative evaluations.

5 Conclusions

We propose MHC-SegNet, a novel and robust encoder–decoder-based multimodal brain tumor segmentation network. Our approach incorporates the SME and TGHE modules within a collaborative encoder, enabling effective capture of long-range dependencies and comprehensive spatial modeling from multiple perspectives. Furthermore, the HFSkip module fuses high-dimensional and low-dimensional features to retain critical information, enhancing segmentation accuracy and consistency. By integrating these innovations, MHC-SegNet achieves best performance on the BraTS2019 dataset with the smallest parameter count among recent methods, demonstrating its potential for clinical applications. In future work, we will further optimize the parameter count and develop networks with even greater clinical dataset, advancing their applicability in medical imaging tasks.

Author Contributions Jiacheng Cao: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Project administration Liyu Ren: Conceptualization, Validation, Writing - Original Draft, Writing - Review & Editing Ao Deng: Validation, Writing - Original Draft, Writing - Review & Editing, Visualization Feng Yu: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing - Review & Editing, Supervision, Funding acquisitionLi Liu: Supervision, Writing - Review & Editing, Investigation, Conceptualization Minghua Jiang: Funding acquisition, Supervision, Investigation, Conceptualization

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Data Availability Data openly available in a public repository. The data that support the findings of this study are openly available in MHC-SegNet at https://github.com/WTU1020-Medical-Segmentation/MHC-SegNet.

Declaration

Conflict of interest The authors declare no competing interests.

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